SCHOOL OF TECHNOLOGY

WOXSEN UNIVERSITY

2026 - 27

CAPSTONE PROJECT – I
REVIEW – I
30TH AND 31ST JULY 2026

Project Title:

REAL-TIME FINANCIAL FRAUD & CREDIT RISK INTELLIGENCE SYSTEM

TEAM MEMBERS

S. No.	Roll No.	Name of the Student	Mobile Number
1	23WU0104039	**K DHEERAJ**	8886668669
2	23WU0104011	**Y. BRIJESH NAIDU**	9666656269
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Mentor Name:

Prof. Dileep Kumar M.

Date of Review : 31.07.2026

SCHOOL OF TECHNOLOGY

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REAL-TIME FINANCIAL FRAUD & CREDIT RISK INTELLIGENCE SYSTEM

Abstract:

Financial fraud presents a critical threat to global banking infrastructure, driving over 40 Billion USD in annual losses worldwide. Traditional rule-based legacy detection systems suffer from severe vulnerabilities, including an inability to adapt to zero-day fraud tactics, failure to uncover collusive multi-account fraud rings, severe class imbalance (~3.5% fraud ratio), and lack of decision explainability required for financial regulatory compliance.

To overcome these limitations, this capstone project proposes a Researcher-Grade End-to-End Machine Learning framework for real-time financial fraud detection and credit risk scoring. The core architecture integrates three specialized paradigms into a unified pipeline: (1) **Big Data Stream Processing** — utilizing Apache Kafka to simulate live, sub-second transaction streams; (2) **Tri-Model ML Fusion** — combining LightGBM (supervised classification), an Autoencoder (unsupervised zero-day anomaly detection), and a Graph Attention Network (PyTorch Geometric GAT) for detecting coordinated multi-account fraud rings; (3) **Adaptive Intelligence** — implementing Concept Drift monitoring (ADWIN) to automatically detect shifting fraud patterns.

The system handles severe class imbalance via SMOTE and tracks all experimental iterations using **MLflow**. The final ensemble outputs a real-time Risk Score per transaction, deployed via a containerized **FastAPI** and **Docker** microservice, with **SHAP** explainability audit trails displayed on a **Streamlit** live dashboard to meet regulatory compliance mandates.

Keywords: Financial Fraud Detection, Big Data Stream Processing (Kafka), Graph Attention Networks (GAT), LightGBM, Concept Drift, FinTech, Explainable AI

Signature of Team Member 3

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Name of Mentor : Prof. Dileep Kumar M.

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INTERNAL EVALUATION SHEET - CAPSTONE PROJECT – I

☑ **Initial Proposal (20%)** ☐ **Mid-Term (30%)** ☐ **Final (50%)**

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S. No.	Criterion	Weightage (%age)	Marks Awarded (out of 100)	Weighted Score
1	Problem Identification	20		
2	Methodology	25		
3	Innovation and Industry Relevance	25		
4	Presentation Clarity	15		
5	Report Quality	15		
	Total	**100**		

Hard Copy of the PPT attached: ☑ **Yes** ☐ **No**

Comments and Feedback Recommendations

Signature of the Mentor